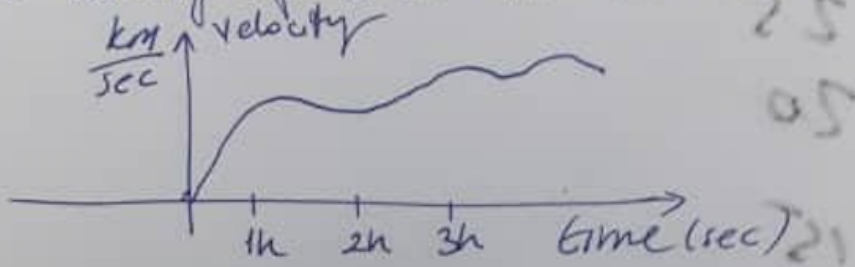
Analog vs Digital W(1).1

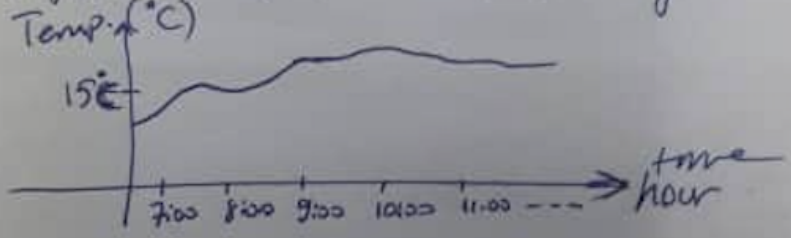
Analog signals: The representation of time varying quantities in a continuous signal.

Examples

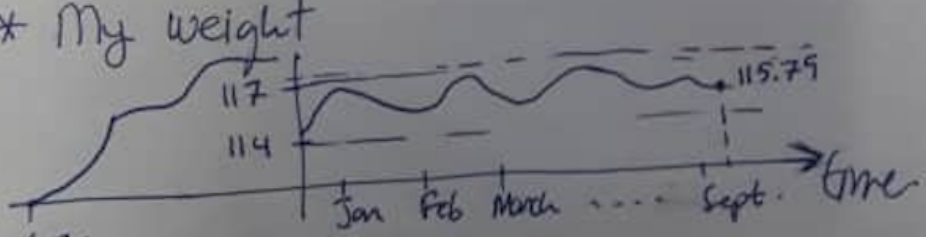
* The velocity of a car in a certain time interval



* The temperature of Istanbul today



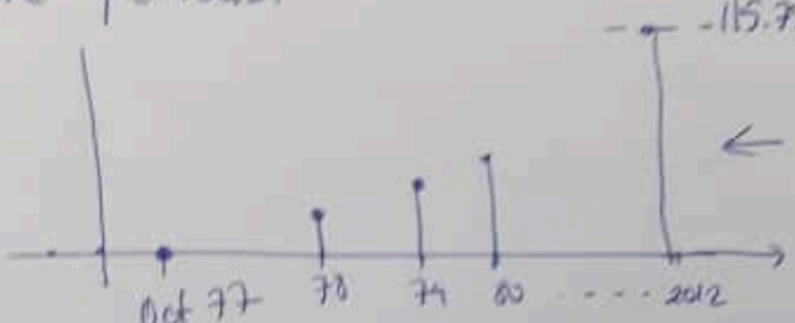
* My weight



Oct 77

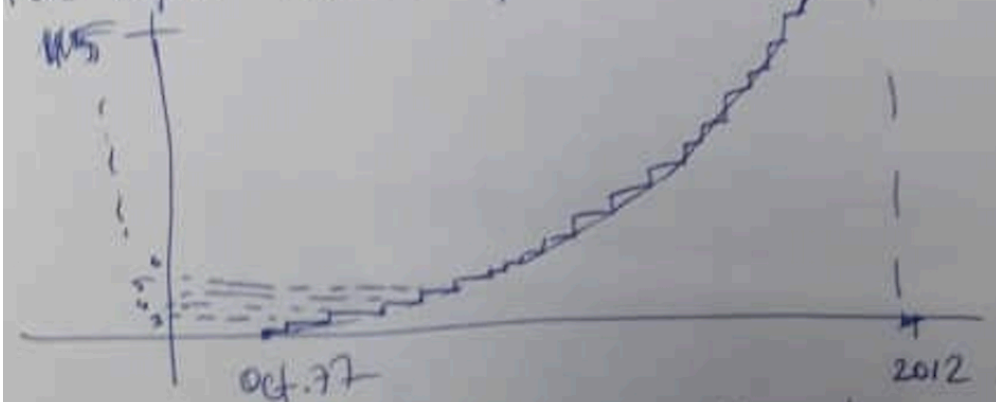
Digital signal: A signal that is a representation of a sequence of discrete values. The signal is sampled and quantized.

What is sampling?: Measure the quantity in certain time periods. (1)



(2)

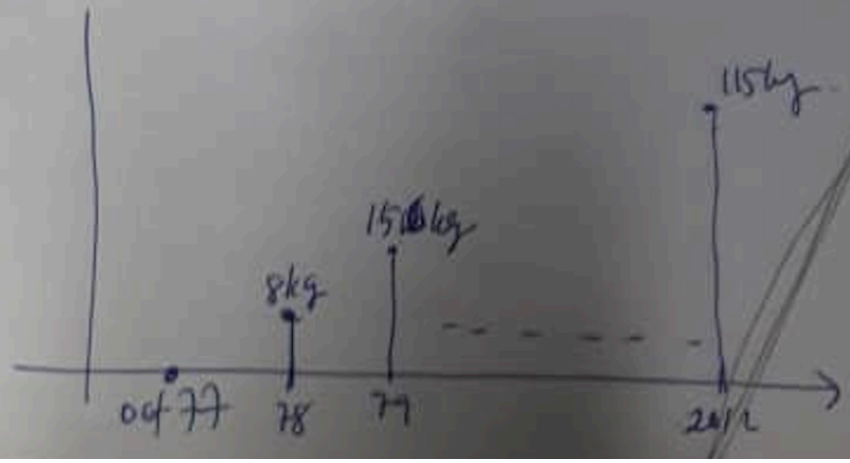
What is quantization: The process of rounding the input values to finite-accurate output values.

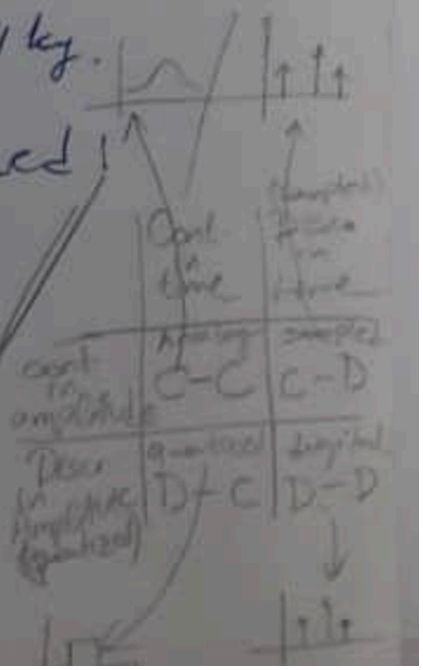


The accuracy: ±0.5kg. (We have an error here)

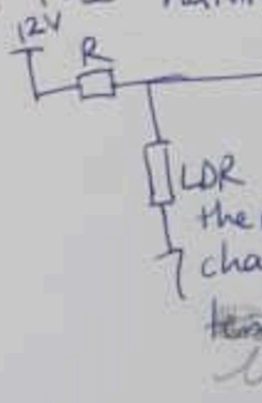
if the weight is 10.6 → round it to 11 kg.

Digital signal: Sampled & quantized!

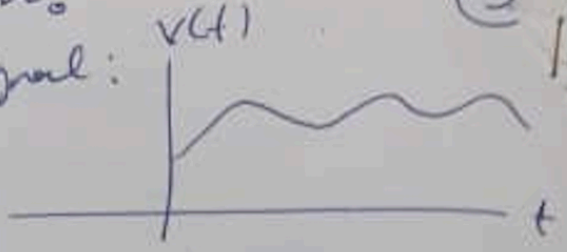




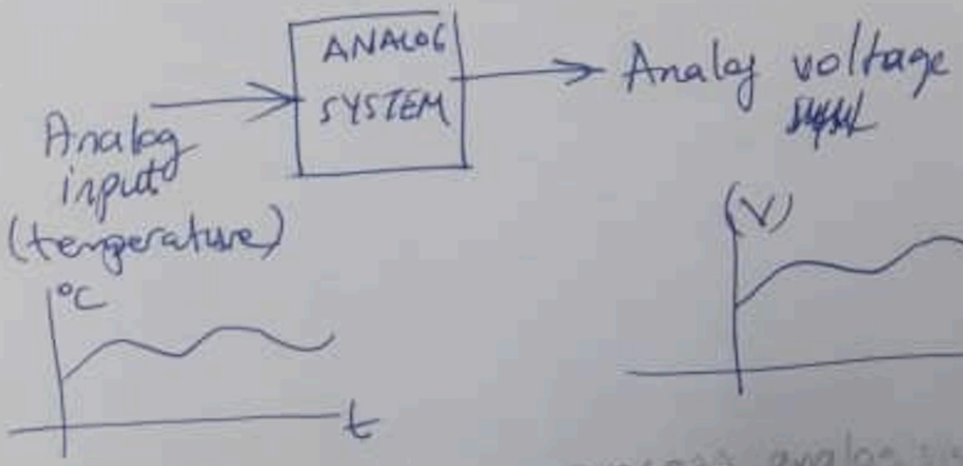
* The nature is an analog system:



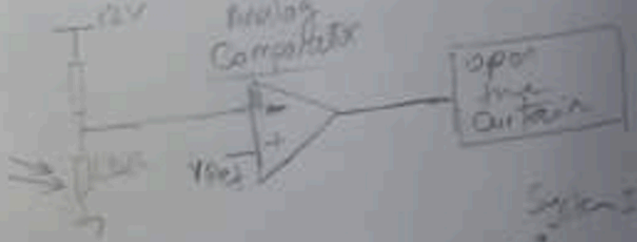
→ The voltage signal:



V(t) follows the ~~temperature~~ light change.



* Analogue systems process analog signals which can take any value within a range.



a simple analog system.

System:

* ~~Disadvantages~~ Analogue:
  + Accurate
  + Uses less bandwidth
  − Cannot be controlled easily.
  − Sensitive to random noise
  − Not flexible/ difficult to manipulate

* Digital Systems:
  + Less expensive
  + More reliable (Not sensitive to noise)
  + ~~Easily~~ controllable
  + Easy to manipulate
  − Sampling error
  − Greater BW required
  − Synchronization requirements. (For some app.)